UNIVERSITY AT ALBANY
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Machine Learning

Lecture 12: Probabilistic Models

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Today

- Lectures after midterm
- Discriminative model vs generative model
- Maximum likelihood estimation
  - Linear regression
- Naïve Bayes models

Lectures after midterm

- Lecture 12: Probabilistic models in supervised learning
  - Maximum likelihood estimation
  - Naïve Bayes model
- Lecture 13-16: Advanced techniques to improve supervised learning
  - Theoretical foundation: Error decomposition
  - Ensemble method (combining multiple classifiers)
  - Kernel method (feature transformation)
  - Non-linear model (neural network)
- Lecture 17-18: Unsupervised learning
  - Clustering
  - Dimension reduction
- Lecture 19: Advanced topics: Decision making and Foundational Models
- Lecture 20: Course review

So far we have learned a lot about ML, but…

- We learned how to
  - Specify a hypothesis class
  - Work out the possible shapes of decision boundaries
  - Train a model by solving an optimization problem
- How did we come up with the hypothesis classes in the first place?
  - We brainstormed… and used
    1. decision trees
    2. linear-classifiers, thresholding a weighted linear combination of features.
  - But how do we know the resulting decision boundaries are appropriate for the problems we hope to solve?

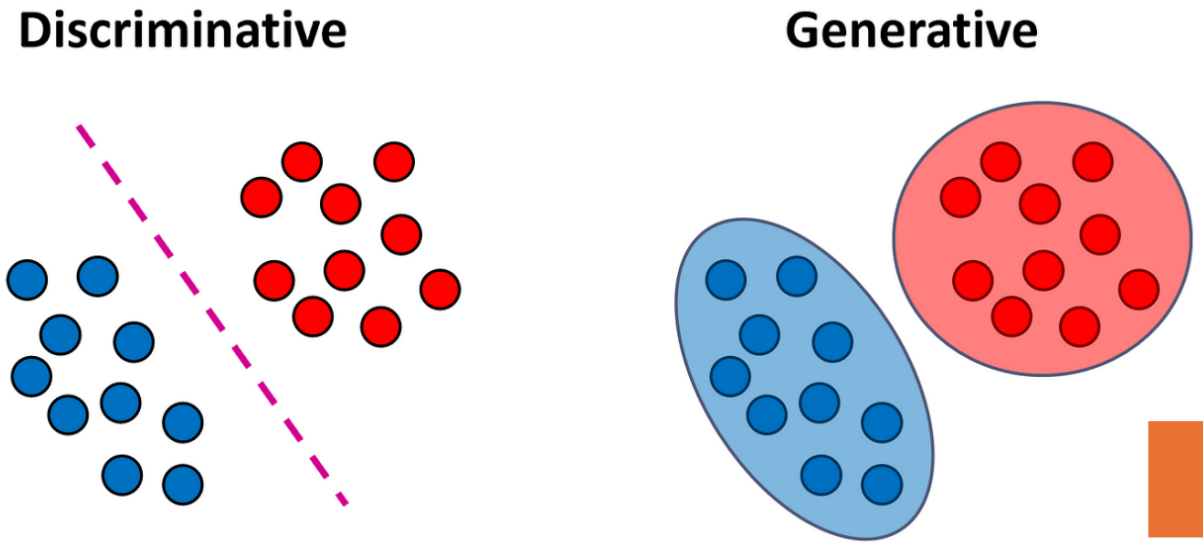
We learned about directly modelling the predictive functions. There is another way… called "Probabilistic modelling"

- We can model how the data are generated in the first place.

  - Model the labeling process via a **conditional distribution** $P(y|x)$.
  - ***(Probabilistic) discriminative model***
    - Non-probabilistic discriminative models:
      - Decision-trees
      - Linear classifiers

  - Model the **joint distribution** $P(x, y)$ by modeling the label distribution $P(y)$ and a generative process $P(x|y)$.
  - ***Generative model.***

Discriminative models vs generative models



	Non-probabilistic	Probabilistic
Discriminative model (how data can be separated?)	Modeling predictive function	Modeling $P(y\|x)$
Generative model (How data are generated?)		Modeling $P(x, y)$ by label distribution $P(y)$ and generative process $P(x\|y)$

Image Credit: Dr. Roi Yehoshua

Probabilistic modelling

- Hard prediction:
  - $h(x) = \text{argmax}_y \, P(y|x)$

- Soft prediction $P(y|x)$
  - Quantifying uncertainty
  - More informative than the score function
  - More interpretable / explainable

How to model $P(y|x)$?

- Bayes rule:
  - $P(y|x) = \frac{P(y)P(x|y)}{P(x)}$
- Key idea:
  - $\text{argmax}_y\ P(y|x) \leftrightarrow \text{argmax}_y\ P(y)P(x|y)$
  - Why?
    - y doesn't depend on $P(x)$.
  - $P(y)$ (distribution of label):
    - Can be estimated by counting labels in training set.
  - $P(x|y)$ (data generating process)

Directly modeling $P(x|y)$ is challenging

Binary vectors, 2^3 rows + binary output $Y \in \{0, 1\}$

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

1. What is the number of data points needed to estimate $p(x|y)$?

2. What happens if there are d binary features?

We need maximum likelihood estimation!

Recap: Maximum likelihood estimation

- Used since Gauss, Laplace, etc….
- Popularized / carefully analyzed by Ronald Fisher.

- Which distribution is more *likely* to have produced the data?

$$\max_{P \in \Pi} f_{\mathrm{Data} \sim P}(\mathrm{Data})$$

Recap: Estimating the mean of Gaussian distribution

- Data

$$X_1, ..., X_n \overset{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$$

- Likelihood:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

- The MLE problem:

$$\hat{\mu} = \arg\max_{\mu \in [0,1]} \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(X_i-\mu)^2}{2\sigma^2}}$$

MLE for linear regression

- $P(y|x)$ is modeled by "Linear Gaussian model"

$$y_i = x_i^T \theta^* + \epsilon_i \quad \text{where } \epsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.}$$

- Data:

$$(x_1, y_1), ..., (x_n, y_n)$$

- In-class exercise: Work out the optimization problem to solve for the MLE for θ^*.

After we fit the MLE, how to make predictions? The idea is to just "Plug-In"

- For classification problems

$$h^*(x) = \max_y p_\theta(y|x) \quad \xrightarrow{\text{Plug in}} \quad \hat{h}(x) = \max_y p_{\hat{\theta}}(y|x)$$

- For regression problems

$$h^*(x) = \mathbb{E}_\theta[y|x] \quad \xrightarrow{\text{Plug in}} \quad \hat{h}(x) = \mathbb{E}_{\hat{\theta}}[y|x]$$

  - Linear Gaussian model: $y \sim N(x^{\mathrm{T}}\hat{\theta}, \sigma^2)$

Recap: Prediction of a probabilistic model

- Step-by-step derivation:
  - $h(x) = \text{argmax}_y P(y|x)$
  - $h(x) = \text{argmax}_y \frac{P(y)P(x|y)}{P(x)}$
  - $h(x) = \text{argmax}_y P(y)P(x|y)$

- Estimating $P(y)$ is easy, but how to estimate $P(x|y)$?

Recap: Direct modeling $P(x|y)$ is hard

Binary vectors, 2^3 rows + binary output $Y \in \{0, 1\}$

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

1. At least 8 data points are needed to determine $p(x|y)$.

2. At least 2^d data points are needed if there are d binary features.

Maximum likelihood estimation (MLE) is one of the ways to model $P(x|y)$.

Naïve Bayes assumption

- Given the class label y, the features are conditional independent of each other.
  - $P(x|y) = \prod_j p(x_j|y)$
  - x_j is the j-th **feature**, not j-th data point!

- How to understand?
  - All features independently affect the label.

Model $p(x|y)$ with naïve Bayes assumption

Binary vectors, 2^3 rows + binary output $Y \in \{0, 1\}$

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Discussion: How many data points are needed to determine $p(x|y)$ given binary output and d binary features?

$$2d$$

Why? When we use the first feature, we ignore all other features…

Naïve Bayes model

- Prediction of probabilistic model:
  - $h(x) = \text{argmax}_y P(y|x)$
  - $h(x) = \text{argmax}_y \frac{P(y)P(x|y)}{P(x)}$
  - $h(x) = \text{argmax}_y P(y)P(x|y)$
  - $h_{\text{Naïve_Bayes}}(x) = \text{argmax}_y P(y) \prod_{j=1}^{d} P(x_j|y)$
- Bayes rule
  - $P(y|x) = \frac{P(y)P(x|y)}{P(x)}$
- Naïve Bayes assumption
  - $P(x|y) = \prod_j p(x_j|y)$

In-class exercise: Naïve Bayes model for email filter

Email ID	Contains "buy"	Contains "cheap"	Contains "free"	Label
1	Yes	Yes	No	Spam
2	No	No	Yes	Not Spam
3	Yes	No	Yes	Spam
4	No	Yes	No	Not Spam
5	Yes	Yes	Yes	Spam
6	No	No	No	Not Spam

- Naïve Bayes model: $h_{\text{Naïve_Bayes}}(x) = \text{argmax}_y P(y) \prod_{j=1}^{d} P(x_j|y)$
- Step 1: calculate $P(y)$
- Step 2: calculate $P(x_j|y)$
- Step 3: prediction on [Yes, No, No]

Naïve Bayes model with continuous variables

- So far we assumed a binomial or discrete distribution for the data given the model ($p(\mathbf{x}^i|y)$)
- However, in many cases the data contains continuous features:
  - Height, weight, Levels of genes in cells, Brain activity

- **Gaussian Naïve Bayes model:**

$$x_i|y \sim \mathcal{N}(\mu_y, \sigma_y^2)$$

Checkpoint: Probabilistic models

- Probabilistic / Generative / Bayesian models are very powerful and interpretable method for modeling the world.
  - Customize ML models for your applications
  - Explicitly model the dependence
- Key task: Modeling $P(x|y)$
  - Maximum likelihood estimation
    - Data are identically and independently distributed (i.i.d.)
  - Naïve Bayes model
    - Features are independent given the label